

Business Intelligence (BI) Dashboards and Visual Analytics in ERP Systems

Submitted to:

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I. Executive Summary

This paper explores the integration of Business Intelligence (BI) dashboards and visual analytics into Enterprise Resource Planning (ERP) systems to enhance data-driven decision-making. ERP systems generate huge amounts of data across finance, supply chain, and human resources, yet often lack advanced analytical capabilities. BI dashboards address this gap by transforming complex ERP data into interactive, real-time visual insights that support key performance monitoring and strategic planning. The study highlights technologies such as Power BI, Tableau, and SAP Analytics Cloud that enable organizations to visualize and interpret data efficiently. Key findings indicate that BI dashboards improve transparency, collaboration, and operational agility while encouraging a data-driven culture. However, challenges include data governance, security risks, integration complexity, and user training needs. The paper recommends ensuring data accuracy and standardization, promoting staff training, and gradually adopting predictive analytics and AI for enhanced forecasting and resource optimization. Lastly, BI-enabled ERP systems empower organizations to make smarter, faster, and more informed decisions in an increasingly competitive digital landscape.

II. Introduction

In the world of business, data-driven organizations create enormous amounts of data points from their Enterprise Resource Planning (ERP) systems, such as finance, human resources (HR), and supply chain. But even though ERP systems are core to operational excellence, they typically lack high-level analytic capabilities required for strategic insights. To fill this void, organizations are connecting Business Intelligence (BI) dashboards and visual analytics applications that convert ERP data labyrinth into valuable, interactive reports, real-time performance tracking, and key performance indicator (KPI) monitoring (Saif, 2024).

The capacity of BI dashboards to display complicated data through interactive interfaces, graphs, and charts makes them relevant in business systems. BI dashboards help managers in identifying KPI, monitoring trends, and more effectively assessing corporate performance by changing huge numbers into visually understandable representations. According to earlier research, integrating BI capabilities into ERP systems enhances organizational competitiveness, operational effectiveness, and strategic planning (Leveraging Business Intelligence Systems for Enhanced Corporate Competitiveness, 2024). Additionally, by facilitating information access across departments and encouraging cooperation and

transparency, visual analytics promotes a data-driven culture. (Implementation of Business Intelligence in ERP Systems, 2020).

The purpose of this study is to explore the role of BI dashboards and visual analytics in upgrading ERP systems. The technology's operation, applications, advantages, difficulties, and potential developments in business settings are the main topics of debate. The report's scope covers an examination of how BI dashboards convert ERP data into actionable insights that facilitate strategic advantage and agile decision-making. This paper shows how BI-driven ERP systems enable businesses to become more competitive, intelligent, and efficient by analyzing existing practices and new developments.

III. Background / Literature Review

Nowadays, many organizations implement Business Intelligence (BI) into Enterprise Resource Planning (ERP) Systems. Large business data from this system is used to show visual insight to executives and managers for easier understanding. The large data is extracted from the systems using the ETL (Extract, Transform, Load) process, then stored in databases and data warehouses. Gonçalves et al. (2023) illustrate that their case study built a data warehouse and Power BI dashboards to merge multiple data sources with the KPI (key performance indicator) required. With the cloud technology in ERP in 2010, systems began embedding analytical capabilities within their offerings. This included integrating tools like Power BI, Tableau for business intelligence, and Qlik for data integration before being used for data visualization (Mhaskey, S. V. 2025).

With nowadays trends, organizations emphasize more agile and intelligent analytics by ERP large data. With algorithms in BI Tools like Power BI, which uses DAX (Data Analysis Expressions), the system can run real-time analytics for agile decision-making and insight. According to Zamil et al., many static reporting tools have been replaced by real-time dashboards for financial purposes. However, according to Zamil et al., there are also drawbacks in automation, which introduce complex issues where the output can be biased. This makes the organization need to be extra careful when using automation and algorithms, as the system can be more problematic because of analytics failure. Advanced technologies nowadays, like ML (Machine Learning) and AI (Artificial Intelligence), also dominate ERP

analytics with forecasting and supply chain optimization techniques (Mhaskey, S. V. 2025). Integrating Internet-of-Things (IoT) into ERP makes dashboards have almost real-time visibility from all the connected devices. Another trend is cloud BI making mobile accessible to the system dashboard (Mhaskey, S. V. 2025).

IV. Analysis and Discussion

Visual analytics and BI dashboard are compulsory for improving the performance of ERP systems. In order to create interactive visual presentations, these dashboards first collect huge amounts of data from ERP modules such as finance, inventories and human resources. Before the data is shown to clients, it must follow an Extract, Transform and Load (ETL) process to ensure correctness and consistency. Businesses may transform difficult statistics into understandable, real-time insights using BI technologies like Oracle BI, SAP Analytics Cloud or Microsoft Power BI. These insights help managers to monitor the operations, assess performance and make strategic choices. (MoldStud, n.d.; BSuperior Systems, 2023).

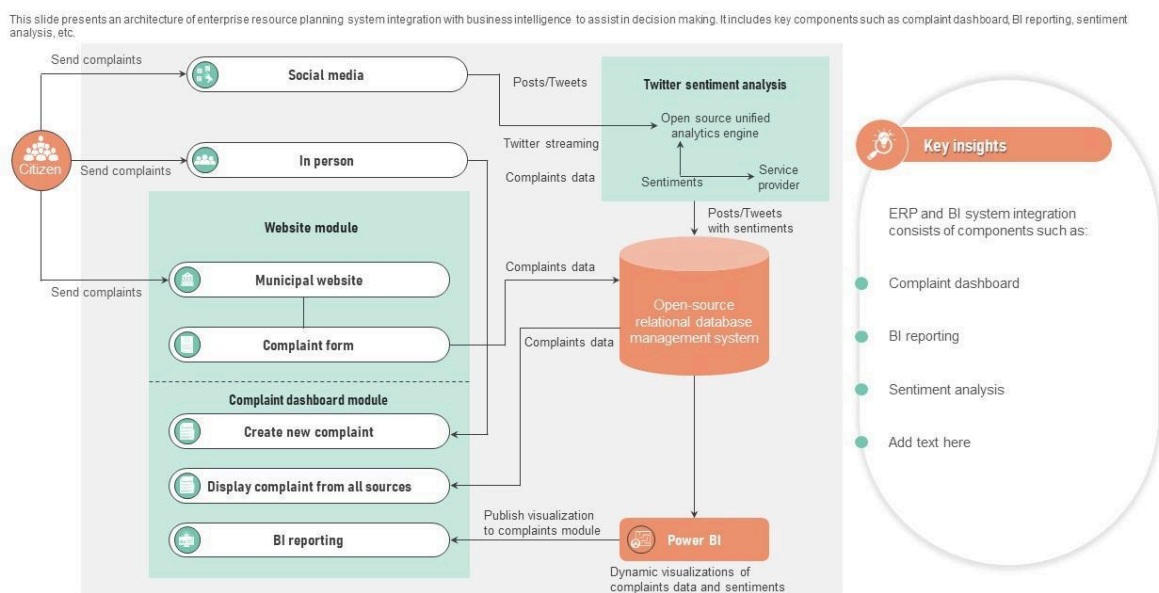


Image 1 : ERP And BI System Integration Design

An organization's whole analytical capability is improved through the integration of BI and ERP systems. Users can interact with data dynamically through BI dashboards rather than relying on static, text-based reports. Managers can find patterns that might not be noticeable in typical ERP reports by using visual analytics to filter data by time, department or product. In addition to supporting a culture of data transparency and accountability across

departments, thus interaction facilitates quicker and better informed decision-making. (BSuperior Systems, 2023; ERP Software Blog, 2023).

Microsoft Power BI's interaction with SAP ERP systems is a significant example of BI dashboard application. With this combination, companies may create personalized dashboards that display real-time financial and operational data. With a few actions, managers may track KPI, keep an eye on sales performance and spot supply chain problems. Real-time ERP data is automatically updated in Power BI dashboards, enabling quick reactions to change in performance or demand. Through accessibility and visual storytelling, this integration increases user engagement and enhances data accurately. (GlobalBSC, 2024; Microsoft Learn, 2024).

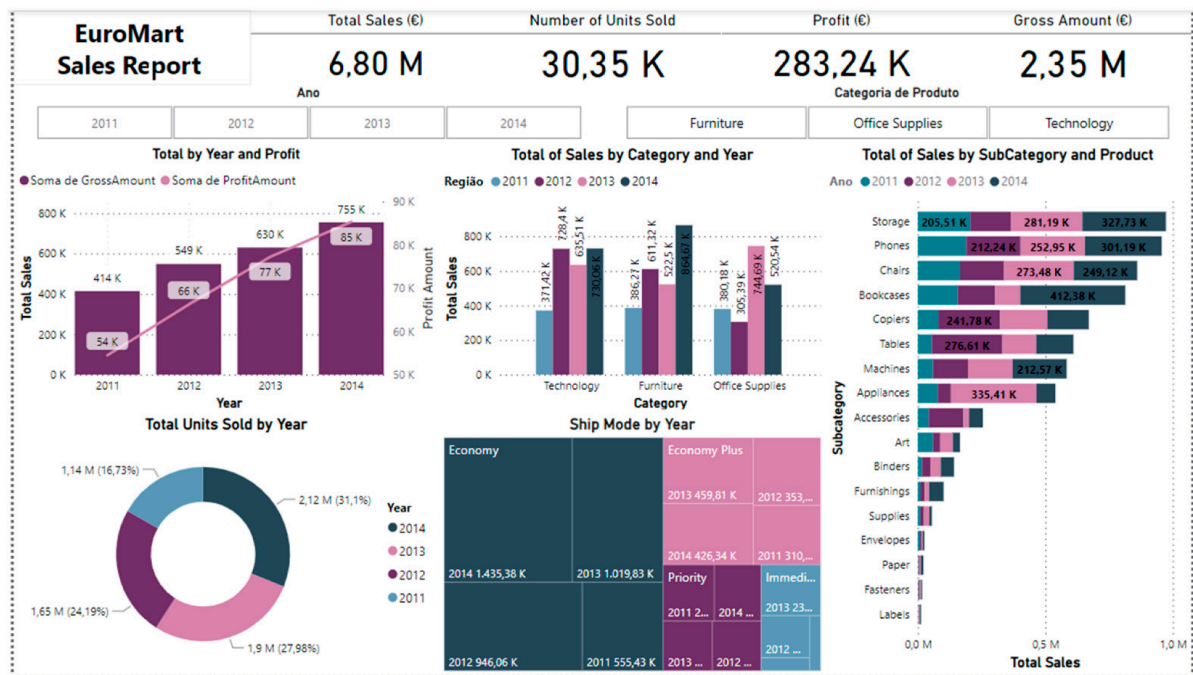


Image 2: Microsoft Power BI ERP dashboard screenshot example

There are many advantages to integrate BI dashboards with ERP systems. By converting huge data sets into clear, comprehensible graphics that emphasize important insights, they accelerate decision-making. Real-time updates from dashboards enable users to make prompt, fact-based business choices. Additionally, because of BI tools are adaptable, lots of departments can create dashboards that are matched to their own terms and requirements. Efficiency, interdepartmental cooperation and increased organisational visibility are all

facilitated by this flexibility. Also, because performance data is transparent and available to all relevant parties stakeholders, BI dashboards help to market accountability. (2WTech, 2024; BSuperior Systems, 2023).

Despite these advantages implementing BI dashboards in ERP environments presents difficulties for enterprises. Consistent formatting and strong data governance are necessary when integrating several data sources. Inaccurate visualizations also may result from human error during data entry or poor data quality. Small and medium-sized businesses may find it expensive to implement BI tools since they need to spend on infrastructure, software and training. Another problem is data security when it comes to cloud-based dashboards that include private company data. IT departments need to keep a balance between security and accessibility. (DataTerrain, 2024; Trigent, 2024; PrescienceDS, 2024).

In future, predictive analytics and artificial intelligence (AI) will be incorporated into BI dashboards. Dashboards will be able to produce automated forecasts and insights with these technologies. Self-service BI technologies are becoming more popular since they let non-technical users make dashboards and reports on their own. Decision-makers will be able to stay connected at any time and from any location with the growth of mobile and cloud-based BI tools, which will further improve flexibility and cooperation. (DynaTech, 2025; Washington Frank, 2025; TEC, 2025).

V. Recommendations

Organizations that integrate BI dashboards in ERP systems should begin by establishing secure information management. Ensuring system data quality, accuracy, and security is essential before visual analytics can provide accurate analytics. Companies should standardize data formats across ERP systems with the implementation of validation procedures to prevent data inconsistency and unauthorized use. With this approach, the dashboards will be more secure from data threats, which makes the system function less effectively.

To maximize the benefits of BI dashboards, organizations must also train their users and promote a data-driven working culture. Employees at all levels should be trained to interpret dashboards and interact with real-time data. This will make all the data-driven work in the

organization more understandable to all employees, reducing ambiguity. Furthermore, using cloud-based BI tools such as Microsoft Power BI or SAP Analytics Cloud can improve data accessibility and collaboration, allowing decision-makers to monitor performance metrics from any location.

Businesses should start using predictive analytics and artificial intelligence (AI) in their BI dashboards to move from handling the problems toward preventing them early. These smart tools can help detect possible issues as soon as possible and manage company resources more efficiently. For smaller businesses, it is best to begin with a step-by-step approach by focusing first on important ERP areas before expanding to other parts of the system.

VI. Conclusion

Visual analytics and BI dashboards are now important parts of contemporary ERP systems. By converting complex statistics into understandable, visually appealing information that facilitates evidence-based decision-making, they close the gap between operational data and strategic insight. Organizations are better equipped to spot inefficiencies, track performance in real time and respond quickly to shifting market conditions because of this connection (Integrating Analytics in Enterprise Systems, 2022). BI dashboards promote accountability, openness and teamwork in businesses through efficient data presentation (Leveraging Business Intelligence Systems for Enhanced Corporate Competitiveness, 2024).

Considering these advantages, complete system optimization still requires addressing issues including data integration, user training and security. The main key is that, in order to fully utilize BI-enabled ERP systems, organizational preparedness, data quality and user engagement are just as crucial as technology itself. ERP systems will continue to develop into more intelligent, self-governing platforms in the future thanks to developments in artificial intelligence, predictive analytics and self-service BI. Future studies might examine how, in an increasingly digital world, these technologies alter organizational performance management and decision-making procedures.

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Appendix / Logbook

GitHub Logbook

Group 11

Backlog

Team capacity

Current iteration

Roadmap

My items

+ New view

Filter by keyword or by field

Todo

0 / 10

Estimate: 0

...

This item hasn't been started

+ Add item

In Progress

0 / 5

Estimate: 0

...

This is actively being worked on

+ Add item

Done

10

Estimate: 0

...

This has been completed

Assignment-1-SECP3744-Sec02 #23

Task Distribution

Iteration 1

Assignment-1-SECP3744-Sec02 #30

Research topic

Iteration 1

Assignment-1-SECP3744-Sec02 #26

Introduction

Iteration 2

Assignment-1-SECP3744-Sec02 #25

Background / Literature Review

Iteration 2

Assignment-1-SECP3744-Sec02 #27

Analysis and Discussion

Iteration 2

Assignment-1-SECP3744-Sec02 #28

Recommendations

Iteration 2

+ Add item